Test 10 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature (Questions 1–15)

1. **Read the passage: *"Ali discovered a series of ancient letters hidden in the library attic. Each letter revealed historical secrets about his town."***
   Question: What did Ali find in the library?
2. **Identify the noun phrases in the sentence: *"Each letter revealed historical secrets."***
3. **Convert to past continuous tense: *"Ali studies the letters carefully."***
4. **Correct the sentence: *"He don't know the author of the letters."***
5. **Identify the verb in: *"The letters contained intriguing mysteries."***
6. **Write a descriptive paragraph (80–100 words) about exploring an old building.**
7. **Compose an email to your teacher explaining a research project you completed.**
8. **Turn into passive voice: *"Ali examined the letters thoroughly."***
9. **Define *metaphor* and use it in an original sentence.**
10. **Summarize the passage in 50 words.**
11. **Convert to a question: *"The students presented their essays on time."***
12. **Write a paragraph describing your favorite hobby (60–80 words).**
13. **Identify the theme in the passage: *"Libraries are treasure troves of forgotten stories."***
14. **Correct punctuation: *"Lets begin reading now before the library closes"***
15. **Explain the difference between autobiography and biography with examples.**

Section B: Mathematics (Questions 16–35)

16. **Solve: $5x + 7 = 32$**
17. **Factorize: $x^2 - 11x + 28$**
18. **Simplify: $(4a^2b)(3ab^3)$**
19. **Solve the system:**
    $x + y = 10$
    $2x - y = 4$
20. **Calculate the area of a triangle with base 12 cm and height 9 cm.**
21. **Solve: $6(2x - 1) = 3x + 15$**
22. **Expand: $(x + 5)(x - 3)$**
23. **Find LCM of 18 and 24.**
24. **A train travels 450 km in 5 hours. Find its average speed.**

25. Solve: $8y/3 + 5 = 13$
26. Find the roots of $x^2 + 7x + 12 = 0$.
27. Find the area of a rectangle with length 14 cm and width 7 cm.
28. Simplify: $(6a^3b^2) \div (2ab)$
29. Probability of getting an odd number on a standard dice.
30. Solve: $4(x - 2) + 3 = 2x + 7$
31. Convert 7/8 into decimal.
32. A polygon has 12 sides. Find each interior angle if regular.
33. Solve: $5(x - 3) = 2x + 9$
34. The ratio of boys to girls is 9:7. If there are 63 boys, how many girls?
35. Simplify: $(7x^2y)(2xy^3)$

Section C: Physics (Questions 36–50)

36. State Newton's First Law of Motion with an example.
37. A 10 N force is applied to a 2 kg object. Find the acceleration.
38. Define uniform and non-uniform motion.
39. A cyclist increases speed from 5 m/s to 15 m/s in 5 seconds. Find acceleration.
40. Define kinetic energy and give an example.
41. A 4 kg object is lifted to 15 m. Calculate its potential energy ($g = 10\ m/s^2$).
42. Explain conservation of energy with an example.
43. Define work and give its SI unit.
44. A 30 N force moves a box 5 m. Find the work done.
45. Differentiate between speed and velocity.
46. State the law of refraction.
47. Give an example of convection in daily life.
48. A 6 Ω resistor has 3 A current. Find the voltage.
49. Explain resistance in terms of material, length, and cross-section.
50. Calculate energy consumed by a 100 W bulb used for 5 hours.

Section D: Chemistry (Questions 51–65)

51. Define a base and give an example.
52. Write the formula for hydrochloric acid.
53. Balance: $Mg + O_2 \rightarrow MgO$
54. Difference between concentrated and diluted solutions.
55. Give examples of homogeneous and heterogeneous mixtures.
56. Electron configuration of nitrogen.
57. Define reduction with example.

58. Write products of metal carbonate decomposition.
59. Is boiling water a chemical or physical change?
60. Contribution of Dmitri Mendeleev to the periodic table.
61. Chemical formula of nitric acid.
62. Define isotope and give an example.
63. Differences between metals and nonmetals.
64. Combustion reaction of propane.
65. Explain acidic, basic, and neutral pH values.

Section E: Biology (Questions 66–80)

66. Photosynthesis equation in plants.
67. Function of chloroplast in plant cells.
68. Difference between eukaryotic and prokaryotic cells.
69. Describe human circulatory system.
70. Define osmosis with example.
71. Function of pancreas in digestion.
72. Two viral diseases affecting humans.
73. Define ecosystem.
74. Difference between herbivores and carnivores.
75. Stages of human excretory system.
76. Types of human muscles.
77. Function of red blood cells.
78. Explain carbon cycle.
79. Role of leaves in photosynthesis.
80. Describe process of pollination.

Section F: Social Studies / History / Geography (Questions 81–90)

81. Define federal government.
82. Three longest rivers in Africa.
83. Causes of World War I.
84. Define global warming.
85. Capital of Canada.
86. Explain human rights.
87. Identify major deserts in Asia.
88. Effects of deforestation.
89. Name three early civilizations in Mesopotamia.

90. Difference between equator and prime meridian.

Section G: Computer / ICT (Questions 91–95)

91. Define algorithm.
92. Two examples of input devices.
93. Functions of spreadsheet software.
94. Difference between software and hardware.
95. Explain importance of data backup.

Section H: Islamiyat / Religious Studies (Questions 96–100)

96. Explain Shahada and its significance.
97. Importance of fasting in Ramadan.
98. Define Hajj and its rituals.
99. Life and teachings of Prophet Musa (AS).
100. Lessons from the story of Prophet Dawood (AS).